FILES ENRYPTION AND DECRYPTION TOOLS

A Project Work Submitted In partial fulfillment of the requirements for the degree

of

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to the

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by

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This to certify that this project work entitled "**FILES ENCRYPTION AND DECRYPTION TOOL"** is a bonafide record of the work done by **Ms. K. HARINI**, **Reg**. **No: C24UG168CCS002** during the academic year 2026 - 2027 in partial fulfillment of the award of the degree of **BACHELOR OF SCIENCE IN COMPUTER SCIENCE WITH CYBER SECURITY.**

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ABSTRACT

ABSTRACT

The **File Encryption and Decryption Tool** is a cybersecurity-based application developed to protect digital files from unauthorized access and ensure the confidentiality of sensitive information. In today's digital environment, users store and share different types of files such as documents, images, academic records, and other personal information. Without proper security, these files may be accessed, modified, or exposed by unauthorized users. Therefore, protecting important digital files is an essential part of information security.

The proposed system provides a simple and effective method for securing files through **encryption and decryption techniques**. Encryption is the process of converting readable data, known as plaintext, into an unreadable format called ciphertext using a cryptographic algorithm and a key. The encrypted file cannot be easily understood without the appropriate decryption key. Decryption is the reverse process, in which the encrypted data is converted back into its original form using the correct key.

The File Encryption and Decryption Tool provides a user-friendly environment in which users can select a file and perform encryption or decryption operations. During encryption, the selected file is processed using a suitable cryptographic technique and stored as an encrypted file. When the original file is required, the authorized user can select the encrypted file and provide the appropriate key to perform the decryption process. The system can also provide suitable messages to indicate whether an operation has been completed successfully or an error has occurred.

The main objective of this project is to provide **confidentiality and controlled access to digital files**. The project also helps users understand important cybersecurity concepts such as cryptography, encryption, decryption, cryptographic keys, file handling, and secure data storage. It can be used for protecting personal files, academic documents, and other confidential information.

INTRODUCTION

CHAPTER 1

1. INTRODUCTION

In the modern digital world, a large amount of information is stored and transferred in the form of digital files. These files may contain personal information, academic documents, business records, financial information, and other confidential data. Protecting such files from unauthorized access is an important part of information security.

Encryption is a security technique used to protect information by converting readable data into an unreadable format called ciphertext. Only an authorized user with the correct key or password can convert the encrypted data back into its original readable form using decryption.

The **File Encryption and Decryption Tool** is developed to provide a simple and secure way to protect digital files. The system allows users to select a file, encrypt it using a suitable cryptographic technique, and save the encrypted file. When required, the authorized user can decrypt the file using the appropriate key or password.

This project demonstrates the practical application of cryptography and provides an understanding of file security, encryption, decryption, key management, and secure data handling.

1.1 Introduction to File Encryption and Decryption Tool

The **File Encryption and Decryption Tool** is a security application designed to protect files from unauthorized access. The main function of the tool is to encrypt a selected file and convert its readable content into a protected format. The encrypted file cannot be normally understood without the required decryption key.

The tool provides two main operations: **Encryption** and **Decryption**. During encryption, the selected file is processed using a cryptographic algorithm and a key. The output is stored as an encrypted file. During decryption, the encrypted file is processed using the correct key to recover the original file.

The proposed tool can be designed with a simple user interface so that users can easily select files and perform encryption or decryption. It can be useful for protecting academic documents, personal files, confidential information, and other important data.

The project also helps students understand how cryptographic techniques can be applied to real-world cybersecurity problems. Proper encryption and secure key management are important for maintaining the confidentiality of digital information.

Main Features

- File selection for encryption and decryption.
- Secure encryption of selected files.
- Decryption using the appropriate key or password.
- Simple and user-friendly interface.
- Error handling for invalid operations.
- Secure storage of encrypted files.
- Protection of confidential information.

1.2 System Specification

System specification defines the hardware and software resources required to develop and run the File Encryption and Decryption Tool. The system should have sufficient processing power, memory, storage, and appropriate software tools to perform file processing and cryptographic operations efficiently.

The requirements are divided into two categories:

1. Hardware Requirements
2. Software Requirements

1.2.1 Hardware Requirements

The hardware requirements specify the physical components needed to develop and execute the project.

Hardware	Minimum Requirement
Processor	Intel Core i3 or equivalent
RAM	4 GB or above
Storage	10 GB free space
Keyboard	Standard keyboard
Mouse	Standard mouse
Display	Standard monitor
Computer/Laptop	Required

1.2.2 Software Requirements

The software requirements specify the operating system, programming environment, development tools, and libraries required for the project.

Software	Requirement
Operating System	Windows / Linux
Programming Language	Python
IDE / Code Editor	VS Code / PyCharm
Cryptography Library	Python Cryptography Library
Database	Optional
Web Browser	Chrome / Firefox
Documentation Tool	MS Word / Google Docs

Python can be used to develop the application because it provides libraries that support cryptographic operations and file handling. A suitable cryptography library can be used to implement secure encryption and decryption rather than creating a custom encryption algorithm.

SYSTEM STUDY

CHAPTER 2

SYSTEM STUDY

2.SYSTEM STUDY

System study is an important stage in the development of the **File Encryption and Decryption Tool**. It involves analyzing the existing methods of file protection and identifying their limitations. Based on these limitations, a proposed system is designed to provide a simple and secure method for protecting digital files. The system study helps to understand the requirements, security needs, advantages, and overall functionality of the proposed application.

2.1 EXISTING SYSTEM

In the existing system, users commonly store their files directly on computers, external storage devices, or cloud platforms. Basic operating system permissions and passwords may be used to restrict access to files and folders. However, these methods do not always protect the actual contents of a file if an unauthorized person obtains access to the storage location.

Some users may also use separate encryption software to protect confidential files. However, such applications can sometimes be complicated for beginners or may provide features that are not necessary for a simple academic project.

In many basic file-storage systems, there is no dedicated process for converting files into an encrypted format before storage or transfer. As a result, sensitive information may be exposed if proper security measures are not applied.

The existing approach mainly depends on normal file permissions, passwords, or third-party applications. Therefore, a simple file encryption and decryption tool can provide a more focused approach to understanding and implementing file security.

2.1.1 DISADVANTAGES OF EXISTING SYSTEM

The major disadvantages of the existing system are:

- Files may be stored without proper encryption.
- Unauthorized users may gain access to sensitive information.
- Basic file permissions may not protect the actual file contents.
- Some encryption applications may be difficult for beginner users.
- Third-party tools may have unnecessary features.
- Secure key management may not be available in simple systems.
- Users may not have a simple interface for both encryption and decryption.
- There is a risk of accidental exposure of confidential files.
- File security may depend heavily on the security of the storage system.

2.2 PROPOSED SYSTEM

The proposed system is a **File Encryption and Decryption Tool** designed to provide a simple method for protecting digital files. The system allows users to select a file and encrypt it using a suitable cryptographic technique. The original file is converted into an encrypted format that cannot be normally understood without the appropriate key.

The system also provides a decryption function to restore the original file. The authorized user can select the encrypted file and provide the correct key or password to perform the decryption process.

The proposed system is designed with a simple and user-friendly interface. It can provide options such as file selection, encryption, decryption, key/password input, output file generation, and status messages.

2.2.1 ADVANTAGES OF PROPOSED SYSTEM

The major advantages of the proposed system are:

- Provides protection for confidential files.
- Converts readable files into encrypted format.
- Allows authorized users to restore files through decryption.
- Provides a simple and user-friendly interface.
- Reduces the risk of unauthorized access.
- Supports secure file storage.
- Helps users understand practical cryptography.
- Provides both encryption and decryption functions in one application.
- Can be extended with advanced security features.
- Suitable for academic and cybersecurity learning purposes.

SYSTEM DESIGN
AND
DEVELOPMENT

CHAPTER 3

3.SYSTEM DESIGN AND DEVELOPMENT

System design and development is an important phase in the development of the File Encryption and Decryption Tool. It describes how the system is structured, how data is given as input, how the encryption and decryption processes are performed, and how the final output is generated. The system is designed to provide a simple and secure environment for protecting digital files.

3.1 FILE DESIGN

File design describes how the input, encrypted, and decrypted files are handled by the system. The user selects a file from the computer using the file-selection option. The selected file is provided as input to the encryption or decryption process.

During encryption, the original file is processed using the selected cryptographic technique and converted into an encrypted file. The encrypted file is stored separately so that the original file is not unnecessarily modified.

During decryption, the encrypted file is selected as input. The system uses the appropriate key or password to restore the original file. The decrypted file is saved with a suitable filename and extension.

The file design should ensure that:

- Original files are handled safely.
- Encrypted files are stored separately.
- Decrypted files can be accessed by authorized users.
- Different file types can be processed.
- File names and extensions are handled correctly.

3.2 INPUT DESIGN

Input design defines the information provided by the user to the system. The input interface should be simple and easy to understand.

The main inputs required are:

- File selection
- Encryption or decryption option
- Password or cryptographic key
- Output file location
- File name, where required

The system should validate the input before processing. If the selected file does not exist, the key is missing, or an invalid operation is selected, an appropriate error message should be displayed.

A proper input design reduces user errors and ensures that the encryption and decryption operations are performed correctly.

3.3 CODE DESIGN

Code design describes the organization of the program used to develop the File Encryption and Decryption Tool. The application is divided into separate functions or modules so that each operation can be easily developed, tested, and maintained.

The major code components include:

- File selection function
- Encryption function
- Decryption function
- Key/password handling function
- File reading and writing function
- Input validation function

A suitable cryptographic library should be used for implementing encryption rather than creating a custom encryption algorithm. The code should also include proper exception handling to prevent unexpected errors during file processing.

3.4 OUTPUT DESIGN

Output design describes the results generated by the system after processing the user's input.After successful encryption, the system displays a confirmation message and provides the location of the encrypted file. Similarly, after successful decryption, the system displays a message indicating that the original file has been restored.

Sample Outputs

Encryption:

"File encrypted successfully."

Decryption:

"File decrypted successfully."

Invalid Key:

"Invalid key or password. Decryption failed."

Invalid File:

"Please select a valid file."

3.5 DATABASE DESIGN

The File Encryption and Decryption Tool can be developed without a database because the main purpose of the application is to process files locally.

If a database is included in the project, it can be used to store non-sensitive information such as user details, file-processing history, timestamps, and operation status. Sensitive encryption keys or passwords should not be stored in plain text.

Example Database Table

Field Name	Data Type	Description
User_ID	Integer	Unique user identification
File_Name	Text	Name of the processed file
Operation	Text	Encryption/Decryption
Date_Time	Date/Time	Operation date and time
Status	Text	Success/Failure

3.6 SYSTEM DEVELOPMENT

System development involves converting the system design into a working application. The development begins with designing the user interface and implementing file-selection functionality. The encryption and decryption functions are then integrated with a suitable cryptographic library.

The system is developed in a modular manner so that each component can be independently tested. Error handling and input validation are also included to improve reliability.

3.6.1 MODULES

The File Encryption and Decryption Tool consists of the following major modules:

1. **User Interface Module**
2. **File Selection Module**
3. **Encryption Module**
4. **Decryption Module**
5. **Key/Password Management Module**
6. **File Management Module**
7. **Error Handling Module**

3.6.2 MODULES DESCRIPTION

1. User Interface Module

This module provides the main interface through which the user interacts with the application. It contains options for selecting files and performing encryption or decryption.

2. File Selection Module

This module allows the user to select the file that needs to be encrypted or decrypted. It also checks whether the selected file is available.

3. Encryption Module

The encryption module converts the original file into an encrypted format using a suitable cryptographic algorithm and key. The encrypted file is then stored securely.

4. Decryption Module

The decryption module converts the encrypted file back into its original form when the correct key or password is provided.

5. Key/Password Management Module

This module handles the key or password required for encryption and decryption. Proper handling of the key is important because losing the required key may prevent recovery of the encrypted file.

6. File Management Module

This module manages reading, writing, saving, and retrieving files during encryption and decryption operations.

7. Error Handling Module

This module identifies invalid inputs and processing errors. It displays appropriate messages when the user selects an invalid file, enters an incorrect key, or encounters another processing problem.

IMPLEMENTATION

CHAPTER 4

4.SYSTEM IMPLEMENTATION

System implementation is the process of converting the designed File Encryption and Decryption Tool into a working application. It includes setting up the development environment, implementing the encryption and decryption functions, managing cryptographic keys, handling files, and managing errors. The implementation is designed to provide secure and reliable file processing.

4.1 IMPLEMENTATION ENVIRONMENT

The implementation environment consists of the hardware and software resources required to develop and execute the application.

The project can be implemented using **Python** because it provides simple file-handling functions and supports reliable cryptographic libraries.

Software Environment

- **Operating System:** Windows / Linux
- **Programming Language:** Python
- **IDE:** Visual Studio Code / PyCharm
- **Cryptography Library:** Python Cryptography Library
- **File Handling:** Python file-handling functions
- **Browser:** Google Chrome / Mozilla Firefox, if required

Hardware Environment

- **Processor:** Intel Core i3 or equivalent
- **RAM:** 4 GB or above
- **Storage:** 10 GB or above
- **Keyboard and Mouse:** Standard input devices

The development environment should provide sufficient memory and storage for processing files and running the application.

4.2 ENCRYPTION PROCESS

The encryption process converts an original readable file into an encrypted format. This prevents unauthorized users from understanding the contents of the file.

Steps:

1. The user selects the file that needs to be protected.
2. The system checks whether the selected file exists.
3. The user provides the required key or password.
4. The application reads the file data.
5. The selected cryptographic algorithm processes the data.
6. The original data is converted into encrypted data.
7. The encrypted data is saved as a new file.
8. The system displays a successful encryption message.

The original file can be retained separately, while the encrypted version is used for secure storage or transfer.

The basic encryption process is:

Select File → Enter Key/Password → Read File → Encrypt Data → Generate Encrypted File → Save Output



4.3 DECRYPTION PROCESS

Decryption is the reverse process of encryption. It converts an encrypted file back into its original readable form using the correct key.

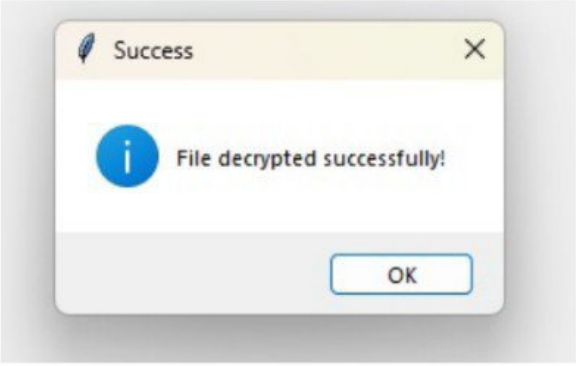
The basic decryption process is:

Select Encrypted File → Enter Key/Password → Read Encrypted Data → Decrypt Data → Generate Original File → Save Output

Steps:

1. The user selects the encrypted file.
2. The system checks whether the file is valid.
3. The user enters the appropriate key or password.
4. The application reads the encrypted data.
5. The cryptographic algorithm processes the encrypted data.
6. The system verifies the required key.
7. The encrypted data is converted back into its original form.
8. The decrypted file is saved.
9. A successful decryption message is displayed.

If an incorrect key is provided, the system should reject the operation rather than producing an invalid output.



4.4 KEY MANAGEMENT

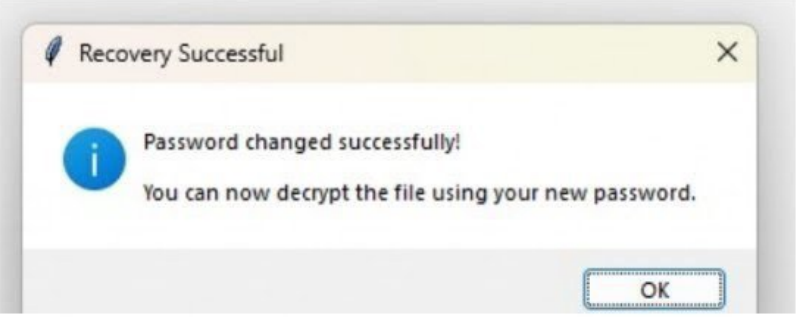
Key management is an important part of the File Encryption and Decryption Tool. The cryptographic key is required to perform encryption and, depending on the algorithm, to decrypt the protected file.

The system should ensure that keys or passwords are handled securely. Keys should not be displayed unnecessarily or stored as plain text. Users should choose strong passwords when password-based encryption is used.

Important key-management practices include:

- Use a strong and appropriate key.
- Do not expose the key to unauthorized users.
- Do not store passwords in plain text.
- Use a trusted cryptographic library for key handling.
- Keep backup or recovery arrangements where appropriate.
- Do not lose the key required for decryption.

Proper key management helps maintain the confidentiality of encrypted files.



4.5 ERROR HANDLING

Error handling is used to identify and manage problems that may occur during file selection, encryption, or decryption. It helps the application operate safely and provides useful information to the user.

The system can handle errors such as:

Error Condition	System Response
File not selected	Display "Please select a file."
File does not exist	Display an appropriate error message.
Incorrect key/password	Reject the decryption operation.
Unsupported file/input	Display an input error message.
File access problem	Display a file access error.
Encryption failure	Display an encryption error message.
Decryption failure	Display a decryption error message.

TESTING

CHAPTER 5

5.SYSTEM TESTING

System testing is an important phase in the development of the File Encryption and Decryption Tool. Testing is performed to verify that the system works according to the specified requirements. It helps identify errors, incorrect inputs, file-processing problems, and other issues before the final implementation.

The File Encryption and Decryption Tool is tested by providing different files, keys, and operations. The encryption process is checked to ensure that the original file is converted into an encrypted format. The decryption process is tested to verify that the original file can be restored using the correct key. Error conditions such as invalid files and incorrect keys are also tested.

5.1 TESTING OBJECTIVES

The main objectives of testing are:

- To verify that the encryption process works correctly.
- To verify that the decryption process restores the original file.
- To ensure that the correct key is required for decryption.
- To check whether invalid inputs are properly handled.
- To identify and correct errors in the application.
- To verify that encrypted files are generated successfully.
- To ensure that decrypted files maintain the required data.
- To check the overall reliability and usability of the system.
- To ensure that the application provides appropriate error messages.

5.2 TEST CASES

The following test cases can be used for the File Encryption and Decryption Tool:

Test Case ID	Test Description	Input	Expected Result	Status
TC01	Select a valid file	Valid file	File selected successfully	Pass
TC02	Encrypt a file	Valid file + correct key	Encrypted file generated	Pass
TC03	Decrypt a file	Encrypted file + correct key	Original file restored	Pass
TC04	Incorrect key	Encrypted file + wrong key	Decryption rejected	Pass
TC05	No file selected	Empty input	Error message displayed	Pass
TC06	Invalid file	Invalid/non-existing file	Appropriate error displayed	Pass
TC07	Encryption failure	Invalid input	Error message displayed	Pass
TC08	Decryption failure	Corrupted/invalid encrypted file	Decryption rejected	Pass

5.3 TESTING METHODS

Different testing methods are used to verify the functionality of the system.

1. Unit Testing

Each individual function of the application is tested separately. For example, file selection, encryption, decryption, and key handling functions are tested independently.

2. Integration Testing

Integration testing verifies whether different modules work correctly together. For example, the file-selection module is tested together with the encryption module to ensure that the selected file is correctly passed for encryption.

3. Functional Testing

Functional testing checks whether the system performs the required operations correctly. Encryption, decryption, file handling, and error messages are tested based on the project requirements.

4. Input Validation Testing

Different types of inputs are provided to the system, including valid files, invalid files, missing keys, and incorrect keys. The system should provide suitable responses for each input.

5. Error Handling Testing

The system is tested under error conditions to ensure that it does not crash and provides understandable error messages to the user.

5.4 TEST RESULTS

After performing the required testing, the File Encryption and Decryption Tool is expected to perform the main operations successfully. Valid files can be selected and encrypted using the appropriate cryptographic process. The encrypted files can be decrypted using the correct key, allowing the original data to be recovered.

The system also handles incorrect keys, invalid files, and missing inputs by displaying appropriate error messages. Testing helps confirm that the major functions of the application work as expected and that the system is suitable for its intended academic purpose

Test Result Summary

Function	Expected Result	Result
File Selection	File selected successfully	Successful
File Encryption	Encrypted file generated	Successful
File Decryption	Original file restored	Successful
Correct Key	Operation completed	Successful
Incorrect Key	Operation rejected	Successful
Invalid File	Error displayed	Successful

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RESULTS AND DISCUSSION

CHAPTER 6

6.RESULTS AND DISCUSSION

The File Encryption and Decryption Tool was successfully developed to protect digital files from unauthorized access. The system allows users to select a file and encrypt it using the Fernet encryption technique. The encrypted file is stored in a separate format. The system also allows the encrypted file to be decrypted using the required recovery key. The original file can be restored after successful decryption. The tool provides suitable messages for successful operations and errors. Overall, the project demonstrates the practical application of encryption and decryption for file security.

6.2 Output Screens

The developed File Encryption and Decryption Tool provides a simple graphical interface for performing file encryption and decryption operations. The output screens demonstrate the selection of files, successful encryption, successful decryption, and error handling. These outputs show that the system performs the required operations correctly and provides clear feedback to the user.

6.3 Discussion

The project demonstrates how cryptography can be used to protect digital files. The encryption process converts the original file into an unreadable encrypted format, while the decryption process restores the original file using the recovery key. The system is simple and easy to use. It also includes error handling for incorrect keys or damaged files.

CONCLUSION
AND
FUTUREENHANCEMENT

CHAPTER 7

7.CONCLUSION AND FUTURE ENHANCEMENTS

7.1 Conclusion

The File Encryption and Decryption Tool was developed to provide a simple and secure method for protecting digital files. The system uses encryption to convert the original file into an unreadable format and decryption to restore the original file using the required recovery key. The project demonstrates the practical application of cryptography, file handling, key generation, and graphical user interface development. The tool is simple to use and helps improve the security and confidentiality of important digital files.

7.2 Future Enhancements

The File Encryption and Decryption Tool can be further improved by adding advanced security features. Future enhancements may include user login and authentication, stronger key management, file integrity verification, multiple encryption algorithms, secure cloud storage, secure file sharing, activity logs, and automatic backup and recovery features. These improvements can make the system more secure, reliable, and suitable for real-world applications.

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APPENDIX

9.APPENDIX – SOURCE CODE

A. SYSTEM ARCHITECTURE

```
from cryptography.fernet import Fernet


key = Fernet.generate_key()

cipher = Fernet(key)


with open("input.txt", "rb") as file:

    data = file.read()


encrypted = cipher.encrypt(data)

with open("encrypted_file.txt", "wb") as file:

    file.write(encrypted)

decrypted = cipher.decrypt(encrypted)
```

```
with open("decrypted_file.txt", "wb") as file:
    file.write(decrypted)

print("File Encryption and Decryption Completed Successfully")
```

B. FILE DATA STRUCTURE

- File Name
- File Type
- File Size
- File Location
- Encryption Status
- Decryption Status
- Encryption Key
- Output File.

C. SAMPLE SOURCE CODE

Python

1. Encryption Code
2. Decryption Code
3. Key Generation Code
4. GUI Code
5. Error Handling Code.

D. SAMPLE INPUT

- Text File

- Image File
- Document File
- PDF File
- Other Supported Files.

E. SAMPLE OUTPUT

- Encrypted File
- Decrypted File
- Encryption Success Message
- Decryption Success Message
- Error Message for Invalid Key or Damaged File